

MARKOV DECISION PROCESS WITH FINITE TIME CRITERIA

FINITE HORIZON DYNAMIC PROGRAMMING FOR HYDRO VALLEYS



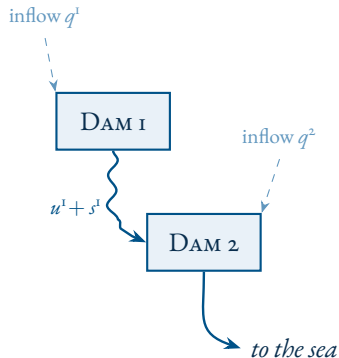
The costs of exact dynamic programming, and an analysis of whether it fails safely

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THE PROBLEM



- Each month, choose how much water each dam turbines.
- Inflows are random; water is sold at every dam it passes.
- State = volume in every dam. Bellman, backwards over 12 months:

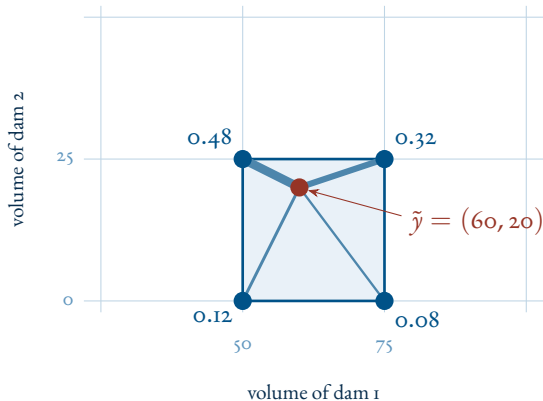
$$\bar{v}_t(x) = \mathbb{E} \left[\min_u \{ c_t(x, u, W) + \bar{v}_{t+1}(f_t(x, u, W)) \} \right]$$

THE QUESTION OF THIS TALK

Exact DP is everyone's reference solution, but it only runs on a grid. *What does the grid cost, and does it fail safely?*

PLATE I. *What dam 1 releases, dam 2 receives.*

ONE TRANSITION, FOUR OUTCOMES



- Next month's volume falls *between* grid points.
- Rounding snaps it to one corner. We **split it** over all four.
- The weights are probabilities; their average is exactly the true point:

$$0.12 (50, 0) + 0.48 (50, 25) + 0.08 (75, 0) + 0.32 (75, 25) = (60, 20)$$

*Not a patch on the model, but an honest
MDP:
exact for Bellman, and **unbiased**.*

PLATE II. The next state falls between grid points.

THE MODEL THAT OVER-PROMISES

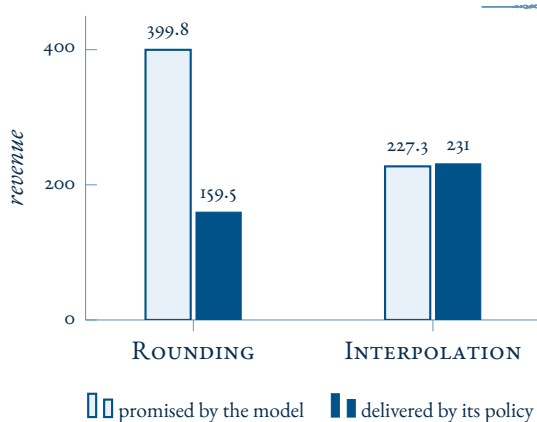


PLATE III. Two dams, coarsest grid ($n = 6$).

- ⚒ Rounding **promises** $2.5\times$ the revenue its policy delivers.
- ⚒ Interpolation very slightly **under**-promises, on every grid tested.

THE SAME ACCURACY NEEDS
121 *vs* **103 041**
interpolation *rounding*
grid points, two dams

an 852-fold smaller state space

WHAT THE POLICY LEARNED

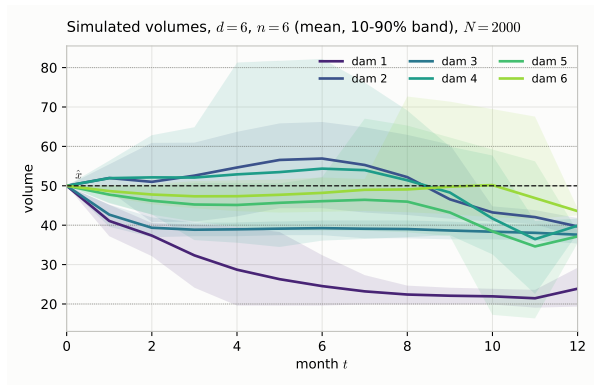


PLATE IV. Six dams, mean simulated volumes.

Given only costs and dynamics, it behaves like a water manager:

- 🏰 **Drains the top dam first:** a unit released there is sold at five dams on the way out.

5.00 vs **0.70**
top of the valley *at the outlet*

- 🏰 Stores through the wet spring, releases in autumn.

THE SIX-DAM VALLEY

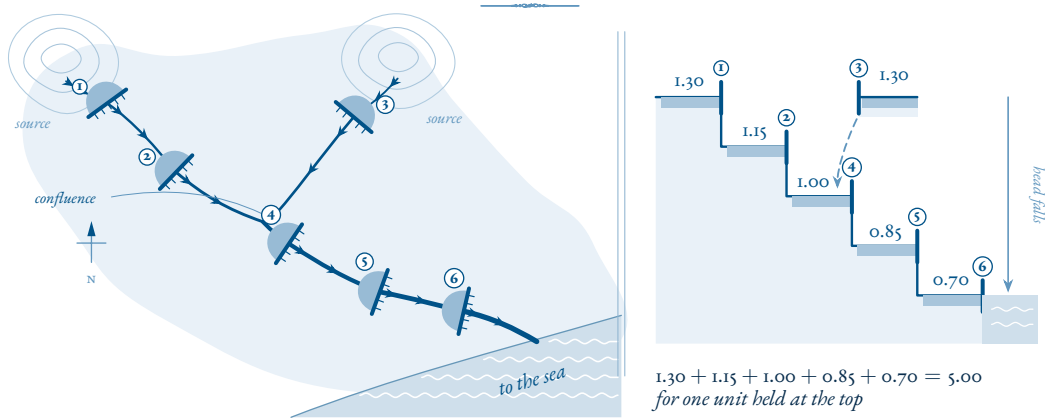


PLATE V. Six dams, two sources, one confluence: the valley from above, and the same dams by height.

THE REAL FRONTIER

SIX DAMS, EXACT BACKWARD INDUCTION

110 minutes, this work

10⁸ minutes, extrapolated
by Carpentier et al. (2018)

- 🚗 They priced a *reference-grade* grid; we accept 7 points per dam.
- 🚗 The limit is **the mesh you accept**, not the number of dams.

AND WHAT THAT MESH COSTS

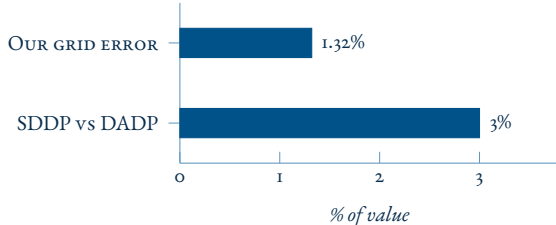


PLATE V. *Six dams. Our grid error is the same order as the disagreement between the two decomposition methods.*

IN CONCLUSION

- I. Splitting each transition over the cell's corners makes the grid model an **honest, unbiased MDP**.
- II. It is $852\times$ **cheaper** than rounding for the same accuracy, and it **errs on the safe side**.
- III. The frontier of exact DP is **the resolution you accept**, not the number of dams.



THANK YOU

APPENDIX · THE BELLMAN RECURSION

Hazard–Decision: the month’s inflow is seen before the release is chosen, so the expectation sits *outside* the minimum. State augmented to (X_t, W_t) , then reduced to E :

$$\bar{v}_t(x) = \mathbb{E} \left[\min_{u \in C_t(x, W_t)} \left\{ c_t(x, u, W_t) + \sum_{\kappa \in \{0,1\}^d} \omega_\kappa(f_t(x, u, W_t)) \bar{v}_{t+1}(g_{k+\kappa}) \right\} \right], \quad \bar{v}_T = \phi.$$

$$\omega_\kappa(y) = \prod_{i=1}^d (\kappa^i \lambda^i + (1 - \kappa^i)(1 - \lambda^i)), \quad \sum_{\kappa} \omega_\kappa = 1, \quad \sum_{\kappa} \omega_\kappa g_{k+\kappa} = y.$$

Cost of the backward pass: $\Theta(T L d (2nm)^d)$ (n^d states, m^d controls, 2^d corners). Rounding saves the 2^d , but the exponential in d remains.

APPENDIX · DOES THE ERROR GROW WITH d ?

d	gap	relative	gap / d
2	6.01	2.68%	3.01 ± 0.28
4	11.12	1.48%	2.78 ± 0.44
6	20.73	1.35%	3.46 ± 0.58

PLATE A. Fixed mesh $\delta = 25$ and fixed $m = 4$; $N = 10^4$.

- It *appeared* to grow: but the 6-dam runs also had a coarser release grid ($m = 4$ vs 10).
- With both grids held fixed, the relative error **falls**.
- Each reservoir adds roughly the same bias, ≈ 3 ; the biases add.

APPENDIX · THE POLICY CONVERGES BEFORE ITS VALUE

d	mesh δ	predicted $\bar{v}_0(x_0)$	realised $\hat{J}_N$	gap
4	25.00	-746.51	-754.51	+8.00
4	14.29	-768.78	-774.58	+5.80
4	9.09	-772.81	-774.99	+2.18
4	6.67	-773.86	-775.42	+1.56

- From $\delta = 14.29$ on, the realised cost moves by 0.84 (0.11%); the prediction still moves by 5.08.
- A coarse grid gives a **near-optimal policy** with an **unreliable price tag**.
- Hence the six-dam value is reported as *at most* -1538.